# An Overview of Technologies in Safety, Security, and Rescue Robotics

Henry Bryant
*Student*
*University of British Columbia*

Aleksandra Vujicic
*Student*
*University of British Columbia*

***Abstract*—The area of Safety, Security, and Rescue Robotics (SSRR) is to promote the advancement of robotics and intelligent systems that can contribute to civilian safety. It focuses on the ethical use of robotics for public safety and security, including applications such as critical infrastructure inspection, nuclear decommissioning, emergency management, humanitarian assistance, and disaster relief [1]. The purpose of this paper is to:**

**· Identify key areas where SSRR technologies are used to enhance public safety and reduce risks to human life,**
**· Evaluate the effectiveness of current commercial SSRR technologies and discuss emerging research trends,**
**· Draw connections between concepts learned in class and their application within SSRR technologies.**

***Index Terms*—SSRR: Safety, Security, and Rescue Robotics**

## I. Introduction

The world shook in 2001 with the attacks on the World Trade Center, commonly referred to as Ground Zero (and will be referred to as such in this paper going forward), which sparked the quick formation of the SSRR Community to address the growing need for advanced robotics to assist in situations where it would be hazardous or impossible for humans to intervene.

The field of SSRR is generally focused into four phases of emergency management whether manmade or natural, comprised of Prevention, Preparation, Response, and Recovery [1]. Prevention is the actions taken to prevent disasters, such as building tidal dams for tsunami protection or providing an alternate means of communication if wireless communication were to go down. Preparation is the actions taken to prepare for an emergency using plans or systems that can help save lives, such as food stocking and using drones to detect early storm warnings. Response is the immediate reaction to an emergency or disaster putting the preparedness plans in action to save lives and prevent more damage, such as active search and rescue during or right after an emergency. Recovery is the action taken to return to a stable or safer situation than before the emergency took place, such as getting financial assistance to fix damage or nuclear decommissioning in hazardous areas.

Some common areas in the SSRR field include "law enforcement, anti-terrorism, nuclear decommissioning, and inspection of critical infrastructure, all phases of emergency management (presentation, preparedness, response, and recovery), and humanitarian assistance and disaster relief" [5]. These areas will be examined in detail with examples of previous (historical) technologies, current technologies, and future (developing) technologies.

## II. What is SSRR?

Safety, Security, and Rescue Robotics (SSRR) refers to the design and application of robotic systems to enhance safety, ensure security, and support rescue operations in hazardous or high-risk environments. These systems use sensing, actuation, and control, to perform critical tasks that minimize human exposure to danger. Equipped with sophisticated sensors such as thermal cameras, LiDAR[1], and gas detectors, SSRR robots can navigate complex terrains, assess environmental conditions, and provide real-time data to operators. They are extensively used in applications like disaster response, industrial safety, surveillance, and search-and-rescue missions. By integrating robust mobility, reliable communication systems, and autonomous or remote-controlled functionalities, SSRR robots ensure high performance in environments where precision, durability, and real-time adaptability are crucial [2].

## III. SSRR Conference Overview

The formulation of the SSRR community as well as the RAS[2]technical committee occurred in 2002, established on February 1st "shortly after the deployment of robots at the 9/11 World Trade Center collapse" [4]. From this followed the development of an annual IEEE conference on SSRR to accelerate the "adoption of robots for homeland security and public safety" [4].

### A. IEEE International Symposium on SSRR

The IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR) is an annual conference focused on the advancement of robotics, automation, intelligent machines, systems and devices that can contribute to civilian safety. The purpose, from the SSRR website, is to "bring together roboticists and practitioners, where academics can present domain-relevant research and engage with agencies, responders, and companies" [5].

As the conference brings together experts from various fields and geographical regions, the papers shown on the IEEE Xplore site in the SSRR field generally vary based on their

[1]LiDAR: Light Detection and Ranging
[2]IEEE Robotics and Automation Society

technical focus areas with some examples being autonomous navigation, multi-robot systems, drones, robots for nuclear decommissioning, or human-robot collaboration. The SSRR conference itself however usually centers around similar core topics each year such as disaster response, hazardous material handling, and search and rescue operations in dangerous and critical environments. The papers focus on all aspects of SSRR, but notably, they focus on the ethical development of robotics for public safety in real-world applications [5]. Notable examples of topics these papers describe fall under law enforcement, anti-terrorism, nuclear decommissioning, infrastructure inspections, emergency management, and disaster relief [5].

The motivation for the conference is to showcase innovations being developed that are based on real user needs. For example, a real-world scenario that would be relevant to the SSRR conference could be based on natural disasters, which would be the problem to focus on for developing innovations in the robotics industry. Common features shown in these robotic innovations have themes regarding multi-robot coordination, autonomous navigation, human-robot interaction, and sensor technologies.

## IV. History of SSRR

There have been a multitude of technological advancements in the area of SSRR which have been captured in research papers over the last two decades. This section will describe some of the key advancements and the History of the SSRR community, first listed as an outline and then covered in detail. I will first cover the key events, such as natural disasters and manmade disasters and what robotics were used in each, and then will cover the robotics in specifics.

### A. *Key Events Timeline*

#### *1) 1999 – RoboCup-Rescue Founded*

An international competition was created (RoboCup, founded in 1997) to develop robotics. Rescue focused on more niche categories below:

- Focused on search & rescue missions and making use of multi-robot coordination for these simulated missions.
- RoboCup itself was designed as an international collaboration to develop robotics to be able to compete against the FIFA world champions in 2050 but has since grown to multiple areas, not including soccer.

#### *2) 2001 – Ground Zero (9/11)*

Key robotics used during and after the wake of the destruction included some of the following:

- Micro-Tracs (Inuktun): Tracked miniature controlled robots
- Micro-VGTV[3] (Inuktun): Variable Geometry Tracked Vehicle with similar uses as the Micro-Tracs
- TALON (Foster-Miller): Tracked ground robot used in rugged environments with a manipulator
- Solem (Foster-Miller: Smaller tracked ground robot used in tighter spaces than TALON
- PackBot (iRobot): Compact tracked robot used in confined and hazardous environments
- UAV prototypes (Not fully deployed due to limitations)

The key problems faced with these early technologies were battery life, reliance on human interaction, and difficulty in wireless communication [1]. This spurred the discussion about the importance of autonomous robotics as an asset going forward.

#### *3) 2002 – SSRR Community and RAS Technical Committee Established*

Shortly after Ground Zero the RAS[4] decided it was necessary to accelerate the adoption of robots used in homeland security and public safety, and created the annual IEEE conference on SSRR to innovate solutions for the growing issue of public safety.

#### *4) 2005 - Hurricane Katrina*

In 2005, a hurricane hit the coast of the United States, passing Florida and worsening due to the warmer waters. This was the first instance of deploying small UAVs for disaster response [1]. The most notable robots used during this disaster were:

- VGTV Xtreme (Inuktun): Track-based robot
- PackBot (iRobot)
- TALON (Foster-Miller, acquired by QinetiQ)
- UAVs deployed by CRASAR

#### *5) 2011 – Fukushima Daiichi, Nuclear Disaster*

Following the Tohoku earthquake on March 11th, 2011, a tsunami caused the meltdown of a nuclear reactor. The most notable robots used in this effort were:

- PackBot (iRobot)
- Honeywell T-Hawk (Westinghouse): Miniature VTOL[5] UAV
- TALON (Foster-Miller, acquired by QinetiQ) in conjunction with bob-cat loaders
- Quince (Chiba Institute of Technology, Tohoku University)

#### *6) 2013 - Oklahoma Tornado*

The 2013 tornado in Oklahoma proved to be one of the first high-profile uses of Drones equipped with LiDAR[6] developed by Texas A&M University. They were designed to fly through areas that were otherwise inaccessible due to weather for storm predictions. We imagine Robin Murphy had a hand in this development as she also leads CRASAR and which is based out of the Texas A&M University [1].

#### *7) 2017/2018 - Hurricane Harvey*

The devastation caused by this hurricane in Texas was followed by flooding, which led to the use of robotics deployed

[3]Variable Geometry Tracked Vehicle

[4]IEEE Robotics and Automation Society

[5]Vertical Takeoff and Landing

[6]Light Detection and Ranging

by CRASAR to aid with search and rescue and damage assessment in flooded areas [8]. While UAVs had a significant role in the search and rescue operations, ground-based and underwater robots were also used, however, these were not as widely documented. The most notable systems deployed were:

- AR200 (Air Robot): Drone with 18lb towing capacity
- Mavic Pro (DJI): LiDAR-equipped Drone

#### *8) **2022 - Hurricane Ian***

During the efforts of the response and recovery during Hurricane Ian, CRASAR deployed UAVs for aerial reconnaissance in areas that were difficult to access due to flooding and infrastructure damage [7]. This was the first large-scale use of drones for technical searches inside structures to assess damage, while also still assisting with search and rescue efforts. The most notable systems used included:

- Mavic 2 Enterprise (DJI): Newer version of Mavic Pro
- Matrice 210 (DJI): Higher payload capacity than Mavic

### *B. Key Advancements in SSRR Timeline*

#### *1) **1990s – Micro-Tracs (Inuktun)***

The Micro-Tracks robot was a robotic track system used for multiple robot assemblies, in this case, a very small track-based tank-like robot. They were designed such that they could fit in small spaces to help with search and rescue, as demonstrated during Ground Zero as a starting point. As mentioned before, this was also part of the three robots used during the initial search and rescue due to their small side. The robot itself was much more portable than the Solem robot, and as such it could be packed and carried easily, as the center of gravity would be very close to the wearer's back [1].

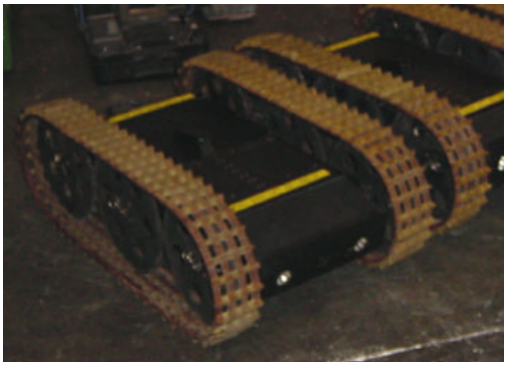


Fig. 1. Micro-Tracs Developed by Inuktun

#### *2) **1990s – Micro-VGTV (Inuktun)***

The Micro-VGTV was a more complex version of the Inuktun robot which had a similar principle in that it was very small and man-packable such that it could be carried easily and maneuvered easily in tight spaces, but it varied in the design because it could adjust its geometry to help with rugged and uneven terrain, including stairs. The tracked wheels can adjust for the situation, and the controller will adjust the orientation of each manipulator based on the required pose of the variable geometry in the robot enabling the robot to climb complex terrain. The technique in general is explained in section V-B later on. The Micro-VGTV encountered issues in the field while testing such as the track belts coming off the guides with no self-centering (which would immobilize the robot), losing traction on uneven surfaces, and debris getting stuck inside the belt drives, which was dealt with on the newer model: the VGTV-Xtreme [12].



Fig. 2. Micro-VGTV Developed by Inuktun

#### *3) $\approx$ **2000 – TALON (Foster-Miller)***

The TALON robot was one of the first semi-autonomous (teleoperated) robots used for hazardous material handling, explosive ordnance disposal (EOD), and reconnaissance in hazardous environments. The TALON had a manipulator that included grippers to manipulate debris in hazardous environments. While it was largely teleoperated, it also included sensor-based feedback for fine manipulation of its robotic arm if the scenario deemed it necessary. The TALON was also developed to maneuver in rugged environments over debris for which was necessary to develop fine-tuned control algorithms to achieve this due to traction and slippage issues. The TALON was one of the first few robots in the early days of SSRR that bridged the gap between remote-controlled and autonomous robots used for SSRR today, as it included a few semi-autonomous features. It was used in various events over the last 2.5 decades, including Ground Zero, Hurricane Katrina, and the Fukushima Daiichi Disaster [1].



Fig. 3. TALON Robot Initially Developed by Foster-Miller

*4) ≈ **2000 – Solem (Foster-Miller)***

The Solem robot was a smaller and more maneuverable version of the TALON robot (first developed by Foster-Miller) with a maneuverable arm which could be equipped with a gripper or camera, which was similar to that of the TALON but more portable. It was able to maneuver into tight spaces controlled remotely by a two-way RF link [9], and as such it was one of the three smallest robots used during the Ground Zero response phase [1]. The gripper was also designed to manipulate debris, as it was a smaller arm designed to be able to lift only light objects [9]. Unfortunately, the Solem robot lost connection deep below the rubble during these events and was never recovered, however, it still provided useful information while it was operational.



Fig. 4. Solem Robot Developed by Foster-Miller

*5) **2001 – PackBot (iRobot, Original)***

The PackBot is a track-based similar to that of a variable-geometry robot, but instead only uses flippers on the front of the robot that would be able to shift around to achieve the differing configurations. It was one of the first widely known robots used in bomb disposal and operation in hazardous environments, and it paved the way for future designs of multiple-geometry SSRR robots used today [1]. The robot included a manipulator that was able to carry about 10kg, which was improved over time to 20kg in the most recent models described by Teledyne [13]. The control methods are explained based on similar robot concepts using pose prediction in section V-A.

*6) **2000s – Micro-VGTV-Xtreme (Inuktun)***

The Micro-VGTV-Xtreme is a newer and more complex model than the Micro-VGTV, where it is more robust in its design enabling more precise operation and improving on the inefficiencies of the previous design as mentioned previously. Another issue that became apparent in the previous design was that it would not be able to control itself if the robot were to flip over, which was fixed in this model. The track slippage due to the track rigidity was solved using a flexible track which would be able to flex under differing surfaces to be able to maintain maximum traction (or as much as possible) to avoid getting stuck [12].



Fig. 5. PackBot Developed by iRobot

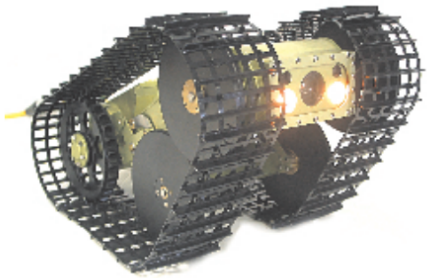


Fig. 6. Micro-VGTV-Xtreme Developed by Inuktun

*7) **2011 – LS3, Precursor to SPOT (Boston Dynamics)***

Development at Boston Dynamics was focused on Quadruped robots (based on the flexibility and maneuverability of a dog), which with the correct manipulation control to maintain stability could easily surpass that of track-based robots in future years. This robot was one of the first advancements towards autonomous quadruped robotics in the SSRR field, which was focused on simulated search and rescue missions in rough terrain. This was vital for improving autonomous navigation from which they would create the newest version of their quadruped robot, SPOT, in 2016 (not covered in this section).




Fig. 7. LS3 Developed by Boston Dynamics

*8) **2010s – Notable Mentions***

While most of the development above has been land-based robotics, there has been a steady improvement in both aerial robotics, as well as underwater robotics. Most of the UAVs mentioned in the timeline of events were useful for the search and rescue events and had autonomous features, such as pre-planned flight trajectories for mapping disasters [7] and locating hard-to-reach areas during search and rescue. However, fully autonomous operation has yet to be achieved, especially in tightly enclosed spaces such as for technical searches in buildings which require specialized pilots to control out of line of sight to be able to assess structural damage for example. Provided that these drones are all different models and achieve the same purpose, this will not be covered and we will focus solely on land-based robotics.

## V. Variable-Geometry Control Concepts

There are a variety of control methods described in research to maintain stability in rough terrain for land-based robotics used in SSRR, categorized into either VGTV's[7] (such as the PackBot which uses additional flippers to modify the robot's shape) or VGSTV's[8] (such as the Micro-VGTV models, which contains a single track and modifies its shape). Before we start this section, we wanted to point out that the difference between a VGTV and a VGSTV does not actually matter in terms of the control solution, because they are both controlled using yaw, pitch, and roll of the maneuverable joint (but in the single tracked case the joints are part of the same track).

### *A. Pose Prediction Control*

Robots such as the iRobot PackBot have flippers to have an additional support mechanism for allowing further stability, which proved useful to improve stability due to careful planning and visual feedback due to manual operation. However, the implementation of autonomous operation would prove to be quite challenging due to the need to predict the robot-terrain interaction which can differ between standard tracked robots and variable-geometry. This prediction works well using heuristic approaches in semi-structured terrain but in the case of rough terrain, it needs to be conservative to avoid accidents which limits the operation of the robot significantly [10]. Normally, heuristic control approaches do not take the joint positions into account; and thus there is a need for a new control method to model the robot-terrain interaction accurately. The scenarios described in this paper are based on the RoboCup Rescue Robot League (RRL) competitions, to get an accurate depiction of unstructured environments.

The pose prediction algorithm is as simple as minimizing the potential energy at any given time. It uses a geometric approach to determine the necessary angle change of the robot flippers to minimize the contact distance and thus the potential energy, where the distance from any point on the robot to the closest surface is found using a Euclidean Signed Distance Field (ESDF). It takes into account the current 3D pose of the robot $(x, y, \phi)$ as well as the joint configurations to calculate the 6D pose $(x, y, z, \phi, \chi, \psi)$ [14]. The 6D pose (and Yaw, Pitch, Roll) is not quite something we have looked at in class, however, it roughly translates to three rotations by the three axes. A transformation matrix is comprised of a rotation matrix where the angle $\theta$ is the same as the yaw angle $\phi$, however, the angle $\alpha$ is the combination of the pitch angle $\chi$ and the roll angle $\psi$ where in the DH parameter case for a single link the pitch angle $\chi = 0$, which makes the angle $\psi = \alpha$. The best example of yaw/pitch/roll is the spherical wrist, where each angle of $\theta_1, \theta_2, \theta_3$ are the yaw, pitch, and roll angles.

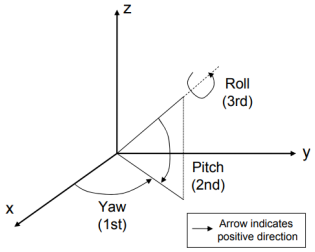



Fig. 8. Yaw, Pitch, and Roll angle Definitions [14]

There are two stages of control operation: the falling stage, and the rotation stage. When the robot is falling, the pose prediction algorithm calculates the first point of contact with the ground by translating the robot along the k-axis as $d^*_{min}$. The ESDF calculates the minimum distance from any point to a surface $d_{min}$ but this could either be the ground or any other close surface. The actual minimum distance is found by translating the robot pose along the k-axis by $d_{min}$ knowing that $d_{min} \leq d^*_{min}$. Since the distance is in the world frame axes, each possible point needs to be converted using the transformation matrix defined by the current robot pose estimation, ${}^{T}W_{C}^{(k)}$[9] before doing the comparison [10]:

$$d_i = \Phi({}^{T}W_{C}^{(k)C} p_i) \tag{1}$$

Note, that $\Phi$ denotes the ESDF function. The pose is found to be valid if there is at least one contact point with the ground and no contact points below the ground, and it can then be used for the rotation stage. The rotation stage determines the least stable axis when the robot pose is in contact with the first point of contact. The least stable axis, denoted as $r$ in the paper, differs depending on the number of contact points ($c$) with the ground [10]:

$$r_1 = (p^{(k)}_{com} - c_1) \times -k_C \tag{2}$$

$$r_2 = c_2 - c_1 \tag{3}$$

Where $p^{(k)}_{com}$ denotes the center of gravity and $\times$ denotes the cross product with the robot -k-axis.

In the case of 3 or more contact points the computation becomes a lot more difficult, using a convex hull of contact

[7] Variable Geometry Tracked Vehicles

[8] Variable Geometry Single Tracked Vehicles

[9] W = World Frame, C = COM Frame of Robot

points (support polygon). Each axis $a_i$ of the support polygon is then tested with the Force-Angle Stability Margin (FASM) to calculate the minimum stability $\beta_{min}$ (which is stable if positive) [10]:

$$r_{3+} = a_i = c_i - c_i \quad \text{If } \beta_i = FASM(a_i) < 0 \tag{4}$$

The rotation stage which is used to provide proper stability is shown visually in the following figure in the paper:

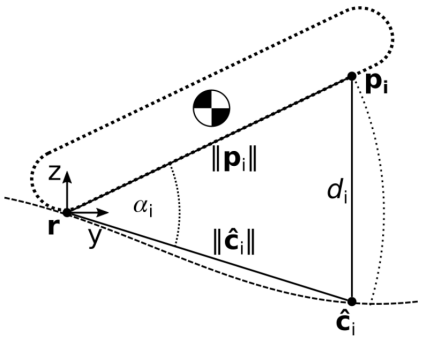



Fig. 9. Rotation Stage Side View [10]

As in the falling stage, they use the distance to the closest surface $d_i$ to predict a possible contact point $\hat{c}_i$ such that rotating the robot (or flipper) by the angle $\alpha_i$ computed using the triangle formed by these three points. The algorithm is briefly explained in the diagram below, which computes the minimum rotation angle $\alpha_i$ along the axis r (mentioned earlier) such that a valid stable contact point is found: As before, since

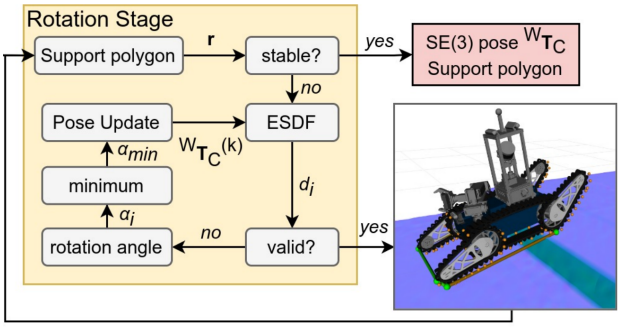



Fig. 10. Rotation Stage Algorithm [10]

the pose in the rotation stage is along the r-axis, they find the pose of the robot $C$ in the $R$ (least stable axis) frame by using the transformation matrices of $R$ in the world frame and $C$ in the world frame. With this pose calculated, each successive pose of the robot in the world frame is calculated by rotating the $R$ world frame by $\alpha_{min}$ by a decreasing step size to avoid oscillations [10]:

$${}^{R}T_{C}^{(k)} = ({}^{W}T_{R})^{-1\,W}T_{C}^{(k)} \tag{5}$$

$${}^{W}T_{C}^{(k+1)} = {}^{W}T_{R}\mathbf{Rot}^{R}T_{C}^{(k)} \tag{6}$$

Overall, the control method described in this paper seemed to have very good results, showcasing exceptional results in both the position error and orientation error for the elevated ramps testing scenario (about twice as efficient) where there are multiple uneven surfaces and gaps/drops on which the pose prediction algorithm would come in very useful. A picture of these ramps is shown below:



Fig. 11. Elevated Ramps Scenario [10]

The data collected for the real and simulated robots used for the elevated ramps is shown below, comparing it to that of the heightmap pose prediction algorithm (HPP) [10]:

TABLE I
ELEVATED RAMPS AVERAGE ERROR

| Model | Position Error (cm) | | Orientation Error (deg) | |
|---|---|---|---|---|
| | *Pose Pred.* | *HPP* | *Pose Pred.* | *HPP* |
| Asterix (Sim) | 2.37 | 7.43 | 6.59 | 12.24 |
| DRZ Telemax (Sim) | 2.08 | 7.97 | 4.57 | 10.54 |
| Asterix (Real) | 3.16 | 6.71 | 5.43 | 11.13 |

Surprisingly, the pose prediction algorithm was still fairly accurate at predicting the position error of the other experiments which are more predictable, and the results were comparable to that of the HPP, which makes this solution a reliable method.

### *B. Zero Moment Point Control*

The Zero moment point control is a control algorithm used for various robots, not necessarily variable geometry, and works by determining a theoretical point where the moment created by all forces has no horizontal components, such that inertia and weight are balanced to ensure stability. As explained in the last section, you can create a support polygon for all contact points of the robot, and if the ZMP goes outside of this support polygon the robot will start to tip over. In the previous section, this was similar, except it only used CoG to determine the stability criterion. In contrast, the ZMP considers both since a robot will generally be in motion and the CoG would not be enough to determine stability. The closer the ZMP gets to the edge of the support polygon of the robot, the more likely the robot will start to tip over, which is an important control aspect to understand. As the ZMP requires accurate ground knowledge to determine stability, this method could easily be combined with the pose prediction control method once the terrain has been mapped to ensure stability, even during motion [11].

## VI. Current Technologies of SSRR

Current commercial applications of SSRR are reshaping industries by enhancing safety and efficiency in challenging environments. These technologies are widely used for tasks such as firefighting, security surveillance, and disaster response. By providing robust solutions for hazardous and high-risk operations, SSRR innovations are becoming indispensable tools in commercial settings, helping businesses protect personnel, and reduce operational downtime.

### A. *Shark Robotics: Collossus*

Shark Robotics, based in France, and has been a leader in the SSRR industry since 2015. Their designs presently focus on firefighting, demining, and security robots, and are actively in use by a number of European organizations. Their heavy firefighting robot, Collossus, was developed in 2017 in collaboration with the Paris Fire Brigade. This modular robot is specifically designed to enhance firefighter safety and efficiency during operations in extreme environments [15]. Colossus gained international recognition during the Notre Dame Cathedral fire on April 15, 2019, where it operated under extreme heat conditions for over 10 hours, and played a critical role in minimizing the risks to human firefighters.

*1) Technical Specifications:*

The Colossus robot is built with an aluminum-welded aeronautical steel chassis, making it highly resistant to heat and water. It can operate for 8-12 hours in extreme conditions without the need for servicing or recharging. Collossus features all-terrain treads capable of climbing stairs up to 30 cm, navigating slopes up to 45°, and traversing various surfaces [16]. It has a payload capacity of up to one ton and can transport equipment or rescue injured individuals. Its water-projection system allows it to project water up to 250 meters, significantly reducing the physical burden on human firefighters.

*2) Key Functionalities:*

Colossus is a modular robot designed to perform multiple tasks, such as extinguishing flames in dangerous zones, transporting equipment, evacuating casualties, and providing real-time environmental data. It is also equipped with advanced sensors, including thermometers and cameras with day/night vision and object-recognition capabilities, which improve situational awareness in low-visibility conditions. Although Collossus is not fully autonomous, its wide range of abilities justify its inclusion in this report. Collossus is controlled remotely by trained firefighters through one of three options: a standard control station, a tablet/smartphone-based system, or a vehicle-mounted station. These systems offer a transmission range of up to 300 meters, ensuring safe operation from a distance.

*3) Additional Features and Applications:*

Shark Robotics currently advertises Colossus to a wide range of industries and boasts nine mission modules that are available with the purchase of Colossus. The nine modules included are:

- Water/Foam Monitor (as pictured in Figure 12)
- 180°Video Turret
- Motorized Bull Bar
- Breathing Air System
- Stretcher for Wounded People
- Equipment Transportation Basket
- Large-Flow PPV
- Arm (with lift capacity of 154lbs)
- Disinfection Kit

Some of the applications that Shark Robotics recommends Colossus be used in includes: warehouse fires, chemical fires, tunnel fires, and car garage fires.



Fig. 12. Shark Robotics: Collossus Heavy Firefighting Robot

### B. *Boston Dynamics: Spot*

Spot, developed by Boston Dynamics, is a versatile quadruped robot designed to navigate complex terrains and perform a variety of tasks across multiple industries. Its agility and adaptability make it suitable for applications ranging from industrial inspections to public safety operations. Boston Dynamics has been developing robots since 1992, and initially began as a spin-off from the MIT Leg Lab. They launched Spot into the commercial space in 2020 and now have over 1500 Spots in operation at various organizations.

*1) Technical Specifications:*

Boston Dynamics' Spot robot is designed to accommodate a variety of payloads, improving its functionality across multiple applications and industries [17]. These payloads support data capture, processing, navigation, and communication tasks. Spot's modular design allows for seamless integration of these payloads without complex setups. They are as follows:

- Spot Arm: Enables manipulation tasks such as opening doors, grasping, dragging, and lifting objects, with a 4K gripper camera for enhanced visibility.
- Spot Cam+ and Spot Cam+IR: Provide advanced imaging capabilities, including high-resolution and infrared cameras, facilitating detailed inspections and data collection.
- Spot GXP (General Expansion Payload): Offers additional power and communication interfaces, including

regulated power outputs (5V, 12V, 24V) and the ability to integrate custom payloads.
- Spot Core I/O: Enhances processing power and connectivity, featuring Ethernet, USB-C, USB-A, and SIM ports, supporting applications requiring substantial data processing and network integration.
- Communication Kits: Provides robust communication solutions for extended operational range and reliability.

*2) Additional Features and Applications:*

Spot has been deployed across a number of different industries, as detailed below:

- Industrial Inspections: Spot can perform detailed inspections in facilities such as oil and gas refineries, chemical plants, and manufacturing sites. When equipped with payloads like Spot Cam+IR (thermal and high-resolution imaging), it detects equipment anomalies, monitors temperature changes, and identifies leaks, ensuring operational safety and preventing costly downtime.
- Energy Sector: Spot can navigate power plants, substations, and offshore oil rigs to inspect pipelines, valves, and electrical systems. Its ability to traverse narrow pathways and handle hazardous conditions reduces the need for human intervention in dangerous areas.
- Public Safety and Emergency Response: In hazardous situations, such as chemical spills or disaster zones, Spot can be deployed to perform reconnaissance, monitor environmental conditions, and locate trapped individuals using thermal imaging. Law enforcement agencies can use Spot for inspecting potentially dangerous areas without risking personnel, such as in bomb disposal scenarios.
- Healthcare and Sanitation: Spot has been used in hospitals in sanitation roles, such as disinfecting large areas using UV-light attachments or spraying systems.

*3) Specific Use Case: Ontario Power Generation*

Additionally, Ontario Power Generation (OPG) has integrated Boston Dynamics' Spot robots into various operations to enhance safety and efficiency. At the Pickering Nuclear Generating Station, Spot is used to inspect hard-to-reach or hazardous areas, reducing risks to human workers. The robot's ability to navigate complex environments, including stairs and obstacles, is controlled remotely via a tablet application. In a world-first achievement, OPG employees successfully replaced a malfunctioning manipulator arm on Spot, marking a significant milestone as the first customer-performed repair of its kind for Boston Dynamics. Beyond nuclear applications, Spot is being tested in hydroelectric stations to conduct inspections and relay data, reducing the need for personnel to travel to remote sites. Future plans include enabling Spot to operate independently at hydro sites, with self-charging capabilities for routine inspections and condition monitoring. By using Spot, OPG plans to improve worker safety, decrease operational costs, and enhance efficiency across its facilities. [18]




Fig. 13. Boston Dynamics: Spot with Arm Attachment

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